

Tutorial 10

COMP 3601

Design and Analysis of Algorithms

Question 1

Solve the following instance of the subset-sum problem by a backtracking technique. Show your state space tree.

$$A = \{3, 4, 5, 6\} \text{ and } d = 13.$$

Question 2

Solve the following instance of the assignment problem by using a branch-and-bound approach.

	Job 1	Job 2	Job 3	Job 4	
$C =$	29	19	17	12	Person <i>a</i>
	32	30	26	28	Person <i>b</i>
	3	21	7	9	Person <i>c</i>
	18	13	10	15	Person <i>d</i>

The cost matrix C represents the assignment costs $C[i, j]$, where persons 1, 2, 3, and 4 are denoted as persons *a*, *b*, *c*, and *d*, respectively.

Question 3

Solve the following instance of the 0/1 knapsack problem by the branch-and-bound technique.

item	weight	value
1	4	\$3
2	3	\$2
3	2	\$4
4	3	\$4

$$W = 6$$

Question 4

Solve the following instance of the traveling salesman problem (TSP) by the branch-and-bound technique.

